

Yingtian Shi

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Education

Georgia Institute of Technology

PhD in Computer Science

Advisor: Prof. Thomas Plötz

Atlanta, GA

Aug 2024 – Present

Expected May 2028

- Research Areas: LLM Agents, Context-Aware AI, Multimodal Sensing, Time Series Learning

Tsinghua University

MS in Computer Science & Technology

Advisor: Prof. Yuanchun Shi

Beijing, China

Aug 2020 – Jun 2023

- Research Areas: Multimodal Interaction, Human Behavior Understanding, Mixed Reality

Tsinghua University

BS in Computer Science & Technology

Beijing, China

Jul 2016 – Jun 2020

Experience

Graduate Research Assistant

AI-CARING, Georgia Institute of Technology

Atlanta, GA

Aug 2025 – Present

- Developed **Rondo** (under review at ICLR '27), combining representation adaptation and variable-length motif modeling for unsupervised temporal structure discovery; achieved the **highest ADR** against SOTA baselines across 6 datasets while **significantly reducing fragmentation**.
- Developed **TRACE** (accepted at IMWUT '26), fusing ambient and wearable sensors with LLM-based temporal reasoning and personalized routine priors; **improved F1 by 11%–33% over the strongest baselines in challenging evaluation settings**, and validated the system in a real-world smart-home deployment, substantially outperforming existing methods.

Research Assistant

PI Lab, Tsinghua University

Beijing, China

May 2025 – Aug 2025

- Built **LLMartini** (under review), an interactive multi-LLM platform, implementing task-aware model recommendation and semantic output fusion for user-controlled comparison and composition; an 18-participant study showed **23.8% lower task completion time, 32.0% lower NASA-TLX workload, and a 25.8-point increase in SUS** over baseline multi-model workflows.

Graduate Research Assistant

Georgia Institute of Technology

Atlanta, GA

Aug 2024 – Apr 2025

- Developed a diabetic foot ulcer detection and segmentation pipeline (IEEE BHI '25) to address poor model performance on underrepresented skin tones in real-world images; evaluated on 3,362 naturalistic foot images from 252 patients, **achieving a 37.7% relative Dice improvement over the strongest baseline**.

Research Assistant

PI Lab, Tsinghua University

Beijing, China

Oct 2023 – Jun 2024

- Developed an expert-in-the-loop visual reasoning framework for construction safety inspection; implemented editable reasoning workflows, visual grounding corrections, and feedback-driven model refinement. After one month of expert collaboration, the system **outperformed Gemini-2.5-Flash by 33.7 percentage points in recall and 16.4 points in F1**.

Visiting Student Researcher

Advised by Prof. James A. Landay, Stanford University

Stanford, CA

Jun 2023 – Sep 2023

- Co-developed **GenieWizard (CHI '25)**, building an LLM-driven agentic missing-API discovery pipeline and integrating it into a VSCode extension with developer-facing interactions; enabled 12 developers to **implement 42% of missing features versus 10% with a standard IDE**.
- Co-developed **ReactGenie (CHI '24)**, translating voice-and-touch commands into executable DSL programs through LLM-based semantic parsing, contextual reference resolution, and composition of application APIs; achieved **90% parsing accuracy on supported user-elicited commands**, enabling multi-step actions from a single multimodal request.

Selected Publications

- TRACE: Temporal Reasoning over Context and Evidence for Activity Recognition in Smart Homes** IMWUT 26
[\[arxiv\]](#) [🔗](#)
Yingtian Shi, Abivishaq Balasubramanian, Jessica Herring, Jiachen Li, Juan Macias Romero, Rosemarie Santa Gonzalez, Varun Mishra, Agata Rozga, Xiang Zhi Tan, Thomas Plötz
- EduGage: Methods and Dataset for Sensor-Based Momentary Assessment of Engagement in Self-Guided Video Learning** IMWUT 26
[\[arxiv\]](#) [🔗](#)
Zikang Leng, Edan Eyal*, Yingtian Shi, Jiaman He, Yaqi Liu, Thomas Plötz*
- Rondo: Unsupervised Discovery of Recurring Temporal Structure** Under review
[\[arxiv\]](#) [🔗](#)
Yingtian Shi, Ankith Chandra*, Thomas Plötz*
- LLMartini: Seamless and Interactive Leveraging of Multiple LLMs through Comparison and Composition** Under review
[\[arxiv\]](#) [🔗](#)
Yingtian Shi, Jinda Yang, Yuhan Wang, Yiwen Yin, Haoyu Li, Kunyu Gao, Chun Yu
- Towards More Equitable Ulcer Recognition Models: A Dataset of Naturalistic Foot Images from People of Color Living with Diabetes** BHI 25
[\[Paper\]](#) [🔗](#)
Cynthia M. Baseman, Yingtian Shi*, Zikang Leng, Yaqi Liu, Gabriel Santamarina, Marcos C. Schechter, Maya Fayfman, Thomas Ploetz, Rosa I. Arriaga*
- GenieWizard: Multimodal App Feature Discovery with Large Language Models** CHI 25
[\[Paper\]](#) [🔗](#)
Jackie (Junrui) Yang, Yingtian Shi, Chris Gu, Zhang Zheng, Anisha Jain, Tianshi Li, Monica S. Lam, James A. Landay
- ReactGenie: A Development Framework for Complex Multimodal Interactions Using Large Language Models** CHI 24
[\[Paper\]](#) [🔗](#)
Jackie (Junrui) Yang, Yingtian Shi, Yuhan Zhang, Karina Li, Daniel Wan Rosli, Anisha Jain, Shuning Zhang, Tianshi Li, Monica S. Lam, James A. Landay
- Understanding In-Situ Programming for Smart Home Automation** IMWUT 23
[\[Paper\]](#) [🔗](#)
Xiaoyi Liu, Yingtian Shi*, Chun Yu, Cheng Gao, Tianao Yang, Chen Liang, Yuanchun Shi*
- (Best Paper Honorable Mention) ConeSpeech: Exploring Directional Speech Interaction for Multi-Person Remote Communication in VR** IEEE VR 23
[\[Paper\]](#) [🔗](#)
Yukang Yan, Haohua Liu, Yingtian Shi, Jingying Wang, Ruici Guo, Zisu Li, Chun Yu, Xuhai Xu, Yuntao Wang, Yuanchun Shi
- Headcross: Exploring head-based crossing selection on head-mounted displays** IMWUT 20
[\[Paper\]](#) [🔗](#)
Yukang Yan, Yingtian Shi*, Chun Yu, Yuanchun Shi*

Technical Skills

Programming Languages: Python, C++, C, C#, Java, SQL, HTML/CSS, TypeScript
Machine Learning: PyTorch, TensorFlow, scikit-learn, OpenCV

Mentored Students

Masters: Abivishaq Balasubramanian, Ankith N Chandra, Yeonsu Shin, Kunyu Gao, Jie He, Guanlin Chen,
Undergraduates: Tianruo Rose Xu, Arya Polavarapu, Luofan Wang, Xuyang Lu, Yuhan Wang, Jinda Yang, Haoyu Li, Yue Liu